

Derivation of $f_{dx}[k]$, $\mathbf{F}_e[k]$, and $\mathbf{H}$

1 Setup

1.1 Plant Equation

$$\gamma \dot{x}(t) = f_{dx}(t) + f_T(t), \quad (1)$$

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + x_D[k]. \quad (2)$$

1.2 Measurement

For a sensor delay of $d = 2$ steps,

$$\delta x_m[k] = \delta x[k-d], \quad \delta x[k] := x_d[k] - x[k]. \quad (3)$$

2 Derivation of the Motion Control Law

This section constructs $f_{dx}[k]$ from (2) so that the closed-loop tracking error follows a first-order pole $\lambda_c \in [0, 1)$. Forced approximations (no alternative) are labelled F1–F6 inline; their residual destinations are summarized in Appendix A.1.

2.1 Design Objective

The requirement is

$$\delta x[k+1] \stackrel{!}{=} \lambda_c \delta x[k]. \quad (4)$$

Substituting (2) into (4) and equating to $\lambda_c \delta x[k] = \lambda_c x_d[k] - \lambda_c x[k]$,

$$a_x[k]\{f_{dx}[k] + f_T[k]\} = x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c)x[k] - x_D[k]. \quad (5)$$

2.2 Idealized Control Law

If all quantities on the right of (5) are instantaneously available, solving for f_{dx} yields

$$f_{dx}[k] = a_x^{-1}[k]\{x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c)x[k]\} - f_T[k] - a_x^{-1}[k]x_D[k]. \quad (6)$$

The unknowns are $a_x[k]$, $x[k]$, $f_T[k]$, $x_D[k]$. The remaining derivation makes (6) implementable using $\delta x_m[k]$, the past control history $f_{dx}[k-i]$, and the EKF estimates $\hat{a}_x[k-j]$, $\hat{x}_D[k-j]$ for $j = 0, \dots, d$.

2.3 State Reconstruction

The sensor provides $x_m[k] = x[k-d] + n_x[k]$, not $x[k]$. Iterating (2) from $k-d$ to k (for $d = 2$, two successive substitutions),

$$x[k-1] = x[k-2] + a_x[k-2]\{f_{dx}[k-2] + f_T[k-2]\} + x_D[k-2], \quad (7)$$

$$x[k] = x[k-1] + a_x[k-1]\{f_{dx}[k-1] + f_T[k-1]\} + x_D[k-1], \quad (8)$$

and summing the two yields the exact identity

$$x[k] = x[k-d] + \sum_{i=1}^d \{a_x[k-i](f_{dx}[k-i] + f_T[k-i]) + x_D[k-i]\}. \quad (9)$$

2.4 Implementation Substitutions

Each unknown in (6) and (9) is replaced by an available quantity:

- **F1** $a_x[k] \mapsto \hat{a}_x[k]$ (EKF posterior).
- **F2** $a_x[k-i] \mapsto \hat{a}_x[k-i]$, $i = 1, \dots, d$ (buffered past posteriors).
- **F3** $x[k-d] \mapsto x_m[k]$ (sensor; the discarded $-n_x[k]$ reappears in the closed-loop noise composite).
- **F4** $f_T[k]$ and $f_T[k-i] \mapsto 0$ (zero-mean white thermal force; unpredictable).
- **F5** $x_D[k] \mapsto \hat{x}_D[k]$ (EKF posterior).
- **F6** $x_D[k-i] \mapsto \hat{x}_D[k-i]$, $i = 1, \dots, d$ (buffered past posteriors).

After F2, F4, F6, the implementable reconstruction (9) becomes

$$\tilde{x}[k] = x_m[k] + \sum_{i=1}^d \{ \hat{a}_x[k-i] f_{dx}[k-i] + \hat{x}_D[k-i] \}. \quad (10)$$

2.5 Final Control Law

Substituting $\hat{a}_x[k]$, $\hat{x}_D[k]$, and $\tilde{x}[k]$ into (6), dropping $f_T[k]$ per F4, and converting $x_m[k] = x_d[k-d] - \delta x_m[k]$,

$$\boxed{\begin{aligned} f_{dx}[k] = & \hat{a}_x^{-1}[k] \{ x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c) x_d[k-d] + (1-\lambda_c) \delta x_m[k] \} \\ & - \hat{a}_x^{-1}[k] (1-\lambda_c) \sum_{i=1}^d \{ \hat{a}_x[k-i] f_{dx}[k-i] + \hat{x}_D[k-i] \} \\ & - \hat{a}_x^{-1}[k] \hat{x}_D[k]. \end{aligned}} \quad (11)$$

The controller buffers $\{\hat{a}_x[k-i]\}_{i=1}^d$ and $\{\hat{x}_D[k-i]\}_{i=1}^d$ in addition to its own past output $\{f_{dx}[k-i]\}_{i=1}^d$. Two design alternatives are discussed in the Appendix: the legacy passive form (§A.2) that collapses the buffered past estimates, and the Meng 2023 limit (§A.3) in which a_x is calibrated and $x_D \equiv 0$.

3 Closed-Loop Tracking Error Dynamics

Define the estimation errors

$$e_a[k] := a_x[k] - \hat{a}_x[k], \quad e_{x_D}[k] := x_D[k] - \hat{x}_D[k]. \quad (12)$$

Since $a_x[k] = \hat{a}_x[k] + e_a[k]$, multiplying (11) by $\hat{a}_x[k]$ gives

$$\begin{aligned} a_x[k] f_{dx}[k] = & e_a[k] f_{dx}[k] + x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c) x_d[k-d] + (1-\lambda_c) \delta x_m[k] \\ & - (1-\lambda_c) \sum_{i=1}^d \{ \hat{a}_x[k-i] f_{dx}[k-i] + \hat{x}_D[k-i] \} - \hat{x}_D[k]. \end{aligned} \quad (13)$$

Substituting (2) and (13) into $\delta x[k+1] = x_d[k+1] - x[k+1]$,

$$\begin{aligned} \delta x[k+1] &= \lambda_c x_d[k] + (1-\lambda_c) x_d[k-d] - x[k] - (1-\lambda_c) \delta x_m[k] \\ &\quad + (1-\lambda_c) \sum_{i=1}^d \{ \hat{a}_x[k-i] f_{dx}[k-i] + \hat{x}_D[k-i] \} \\ &\quad - e_a[k] f_{dx}[k] - e_{x_D}[k] - a_x[k] f_T[k]. \end{aligned} \quad (14)$$

Substituting (3) for $\delta x_m[k]$ and (9) for $x[k]$, the $x_d[k-d]$ terms cancel; the remaining $\lambda_c(x_d[k] - x[k-d])$ returns to $\lambda_c \delta x[k]$ through (9):

$$\begin{aligned} \delta x[k+1] &= \lambda_c \delta x[k] - e_a[k] f_{dx}[k] - e_{x_D}[k] - a_x[k] f_T[k] \\ &\quad - (1-\lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] \\ &\quad - (1-\lambda_c) \sum_{i=1}^d (x_D[k-i] - \hat{x}_D[k-i]) - (1-\lambda_c) \sum_{i=1}^d a_x[k-i] f_T[k-i]. \end{aligned} \quad (15)$$

Applying (12) at steps $k-i$,

$$\begin{aligned} \delta x[k+1] &= \lambda_c \delta x[k] - e_a[k] f_{dx}[k] - e_{x_D}[k] \\ &\quad - (1-\lambda_c) \sum_{i=1}^d \{ e_a[k-i] f_{dx}[k-i] + e_{x_D}[k-i] \} - \varepsilon_T[k], \end{aligned} \quad (16)$$

where

$$\varepsilon_T[k] := a_x[k] f_T[k] + (1-\lambda_c) \sum_{i=1}^d a_x[k-i] f_T[k-i]. \quad (17)$$

Every residual in (16) is an estimation error (e_a , e_{x_D}) or a thermal forcing (f_T); no Taylor expansion of a_x is involved, hence no model-truncation residual appears.

For brevity, define

$$\mathcal{F}_1[k] := \sum_{i=1}^d f_{dx}[k-i], \quad \mathcal{F}_2[k] := \sum_{i=1}^d i f_{dx}[k-i]. \quad (18)$$

4 State Vector and Output Matrix

Estimation of $\hat{a}_x[k]$, $\delta \hat{x}[k]$, and $\hat{x}_D[k]$ is required to update (28). Due to the d -step measurement delay, three variables $\delta x_1[k] = \delta x[k-2]$, $\delta x_2[k] = \delta x[k-1]$, $\delta x_3[k] = \delta x[k]$ model the motion error, and x_D and a_x are modeled as second-order processes. The augmented state vector is

$$\mathbf{x}[k] = [\delta x_1, \delta x_2, \delta x_3, x_D, \delta x_D, a_x, \delta a_x]^T[k], \quad (19)$$

whose evolution is given by

$$\begin{cases} \delta x_1[k+1] = \delta x_2[k] \\ \delta x_2[k+1] = \delta x_3[k] \\ \delta x_3[k+1] = \lambda_c \delta x_3[k] - f_{dx}[k] e_a[k] - e_{x_D}[k] - (1-\lambda_c) \sum_{i=1}^d x_D[k-i] - \varepsilon_T[k] + \eta_R[k] \\ x_D[k+1] = x_D[k] + \delta x_D[k] \\ \delta x_D[k+1] = \delta x_D[k] \\ a_x[k+1] = a_x[k] + \delta a_x[k] \\ \delta a_x[k+1] = \delta a_x[k] \end{cases} \quad (20)$$

The two measurements $y_1[k] = \delta x_m[k]$ and $y_2[k] = a_{xm}[k]$ both inherit the d -step delay; in particular $y_1[k] = \delta x_1[k]$ and $a_{xm}[k] \approx a_x[k-d] + n_a[k]$, where $n_a[k]$ is the intrinsic measurement noise. By the second-order process for a_x in (20),

$$a_x[k-d] = a_x[k] - d \delta a_x[k], \quad (21)$$

so that $a_{xm}[k] \approx a_x[k] - d \delta a_x[k] + n_a[k]$, yielding

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -d \end{bmatrix}. \quad (22)$$

5 Closed-Loop Estimator and Error Dynamics

Through the feedback of two estimation errors $e_x[k] = \delta x_m[k] - \hat{\delta x}_1[k]$ and $e_{a_x}[k] = a_{xm}[k] - \hat{a}_x[k]$, the closed-loop estimator is designed as

$$\begin{cases} \delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_x[k] + \ell_{12}[k] e_{a_x}[k] \\ \delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_x[k] + \ell_{22}[k] e_{a_x}[k] \\ \delta \hat{x}_3[k+1] = \lambda_c \delta \hat{x}_3[k] - d(1-\lambda_c) \hat{x}_D[k] + \frac{d(d+1)}{2} (1-\lambda_c) \delta \hat{x}_D[k] \\ \quad + (1-\lambda_c) \mathcal{F}_2[k] \delta \hat{a}_x[k] + \ell_{31}[k] e_x[k] + \ell_{32}[k] e_{a_x}[k] \\ \hat{x}_D[k+1] = \hat{x}_D[k] + \delta \hat{x}_D[k] + \ell_{41}[k] e_x[k] + \ell_{42}[k] e_{a_x}[k] \\ \delta \hat{x}_D[k+1] = \delta \hat{x}_D[k] + \ell_{51}[k] e_x[k] + \ell_{52}[k] e_{a_x}[k] \\ \hat{a}_x[k+1] = \hat{a}_x[k] + \delta \hat{a}_x[k] + \ell_{61}[k] e_x[k] + \ell_{62}[k] e_{a_x}[k] \\ \delta \hat{a}_x[k+1] = \delta \hat{a}_x[k] + \ell_{71}[k] e_x[k] + \ell_{72}[k] e_{a_x}[k] \end{cases} \quad (23)$$

where $\ell_{ij}[k]$ are the entries of the 7×2 estimator feedback matrix $\mathbf{L}[k]$. The Row-3 predict terms in $\hat{x}_D$, $\delta \hat{x}_D$, and $\delta \hat{a}_x$ are chosen so that subtracting (23) from (20) reproduces the error dynamics (24).

The dynamics of estimation errors $\mathbf{e}[k] = \mathbf{x}[k] - \hat{\mathbf{x}}[k]$ is obtained by subtracting (23) from (20):

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k], \quad (24)$$

where $\mathbf{q}[k]$ is the process noise and $\mathbf{F}_e[k]$ is the state-transition matrix of the error dynamics given in §6.

6 State Transition Matrix

By the second-order process for x_D in (20), each past value $x_D[k-i] = x_D[k] - i \delta x_D[k]$. Summing over $i = 1, \dots, d$ and using $\sum_{i=1}^d i = d(d+1)/2$,

$$\sum_{i=1}^d x_D[k-i] = d x_D[k] - \frac{d(d+1)}{2} \delta x_D[k]. \quad (25)$$

Substituting (25) into Row 3 of (20) and absorbing the state-dependent part of η_R (via $\mathcal{F}_1, \mathcal{F}_2$ from (18)),

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -(1+d(1-\lambda_c)) & \frac{d(d+1)}{2}(1-\lambda_c) & -f_{dx}[k] - (1-\lambda_c)\mathcal{F}_1[k] & (1-\lambda_c)\mathcal{F}_2[k] \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \quad (26)$$

A Assumptions and Design Alternatives

A.1 Catalog of Forced Approximations

Tag	Substitution	Reason
F1	$a_x[k] \mapsto \hat{a}_x[k]$	True a_x is unobservable; only the EKF posterior is available.
F2	$a_x[k-i] \mapsto \hat{a}_x[k-i]$	Past true a_x is unobservable; buffered past posteriors used.
F3	$x[k-d] \mapsto x_m[k]$ (drop n_x)	Sensor noise n_x is unobservable.
F4	$f_T[\cdot] \mapsto 0$	Zero-mean white thermal force; no predictor exists.
F5	$x_D[k] \mapsto \hat{x}_D[k]$	True x_D is unobservable; only the EKF posterior is available.
F6	$x_D[k-i] \mapsto \hat{x}_D[k-i]$	Past true x_D is unobservable; buffered past posteriors used.

All six are forced: the substituted quantity is either unobservable (F1, F2, F5, F6) or zero-mean white noise without a predictor (F3, F4). The single design parameter $\lambda_c \in (0, 1)$ controls the closed-loop convergence rate versus noise injection trade-off.

A.2 Legacy Passive Form

Collapsing the buffered past estimates in (11) to their current values,

$$\hat{a}_x[k-i] \mapsto \hat{a}_x[k], \quad \hat{x}_D[k-i] \mapsto 0 \quad (i \geq 1), \quad (27)$$

the control law simplifies to

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \{ x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c) x_d[k-d] + (1-\lambda_c) \delta x_m[k] \} - (1-\lambda_c) \sum_{i=1}^d f_{dx}[k-i] - \hat{a}_x^{-1}[k] \hat{x}_D[k].} \quad (28)$$

This is the form used in §3–§5. Under (28) the closed-loop (16) carries the raw past disturbance $-(1-\lambda_c) \sum_i x_D[k-i]$, and the $\hat{a}_x[k-i]$ contributions reduce, via the Taylor expansion $a_x[k-i] = \hat{a}_x[k] + e_a[k] - i \delta a_x[k]$, to the η_R residuals carrying $e_a \mathcal{F}_1$ and $\delta a_x \mathcal{F}_2$. The Row-3 predict in (23) therefore retains $-d(1-\lambda_c) \hat{x}_D + \frac{d(d+1)}{2} (1-\lambda_c) \delta \hat{x}_D + (1-\lambda_c) \mathcal{F}_2 \delta \hat{a}_x$ so that subtraction reproduces the same $\mathbf{F}_e[k]$ (26).

The full form (11), by contrast, replaces $-(1-\lambda_c) \sum_i x_D[k-i]$ by the estimation-error sum $-(1-\lambda_c) \sum_i e_{x_D}[k-i]$ in (16), eliminating raw-disturbance leakage; the same $\mathbf{F}_e[k]$ is recovered because the error-dynamics matrix is invariant under the controller–estimator split of responsibility.

A.3 Relation to Meng 2023 Eq. 17

Setting $x_D \equiv 0$, $\hat{x}_D \equiv 0$, and treating a_x as a known calibrated constant $\Delta t/\gamma$ (i.e. skipping F1, F2, F5, F6), the passive form (28) reduces to

$$f_{dx}[k] = (\gamma/\Delta t) \{x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c) x_d[k-d] + (1-\lambda_c) \delta x_m[k]\} - (1-\lambda_c) \sum_{i=1}^d f_{dx}[k-i], \quad (29)$$

which is Eq. 17 of Meng 2023. The present (11) is a strict generalization that estimates a_x online (F1, F2) and incorporates a position-dependent disturbance compensator (F5, F6).